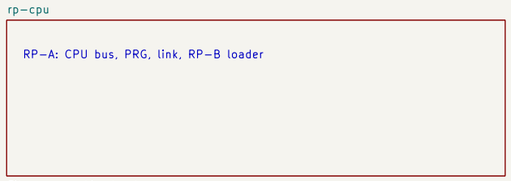


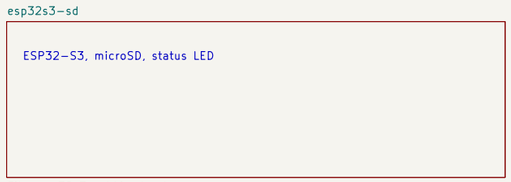
File: edge–cic.kicad_sch



File: rp–cpu.kicad_sch



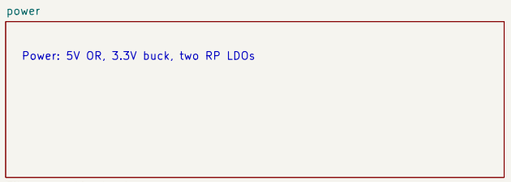
File: rp–ppu.kicad_sch



File: esp32s3–sd.kicad_sch



File: usb–uart.kicad_sch



File: power.kicad_sch

FujiNet for the Nintendo Entertainment System – Rev1 (two RP2354Bs, no SRAM)

RP–A (U1) sits on the CPU side of the 5V cart bus through its 5V–tolerant pads: CPU A0–A14, D0–D7, M2, R/W, /ROMSEL in, /IRQ out, PPU A12 in (MMC3 IRQ). It serves PRG from RAM (images up to 384K) or from its in–package flash through QMI address translation (512K), plus $5000–$7FFF, the loader and the vectors, and runs the mappers.
RP–B (U2) sits on the PPU side: PPU A0–A13, D0–D7, /RD, /WR in; CIRAM A10 and /CE out. It serves CHR, CHR–RAM and four–screen nametables from RAM and has no firmware of its own: RP–A loads it over SWD (GP44–46) at every boot. RP–A forwards every mapper write to RP–B over an 8–bit link (LD0–7, LCLK, LFRM, LACK back).
Each RP's whole supply (IOVDD, core regulator input) runs from its own AP2112K off +5V (+3V3_RP, +3V3_RPB), so the pads are powered whenever the console bus is. PWR_OK = 0.6 x CONS_5V (2.2k / 3.3k) into both RPs: the data lines and CIRAM stay off the bus until the console is on; RP–B also reads it on its ADC.
ESP32–S3–WROOM–1–N16R8 runs FujiNet (fujiversal–nes) and is the USB host of RP–A (CDC, VID 0xCafe); S3 IO4/IO5 force RP–A RUN/QSPI_SS so the S3 can re–flash RP–A over PICOBOOT.
USB–C: power + CP2102N UART bridge for flashing the S3. microSD on S3 SPI. CIClone (ATtiny13A) on the CIC fingers, DNP until its firmware exists.
Edge: 72 pins, 2.50 mm pitch, 1–36 on the label side, pin 1 at the RIGHT seen from the label side with the fingers down (nesdev Cartridge connector; NES–EWROM–01 geometry). EXP, SYSTEM CLK and PPU /A13 unconnected.

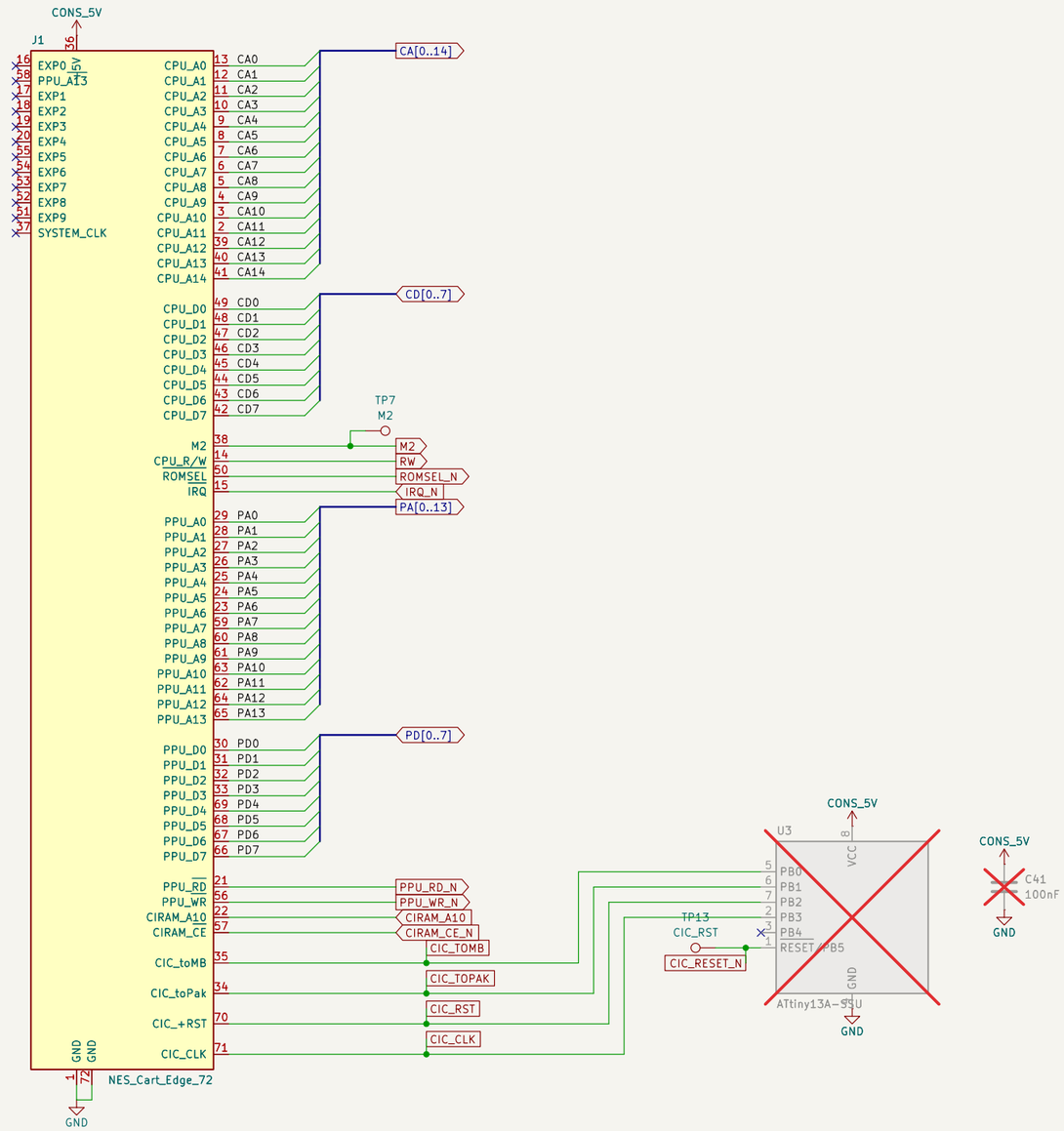
Generated by tools/gen_sch.py
FujiNet

Sheet: /
File: FujiNet–NES–Rev1.kicad_sch

Title: FujiNet NES Rev1

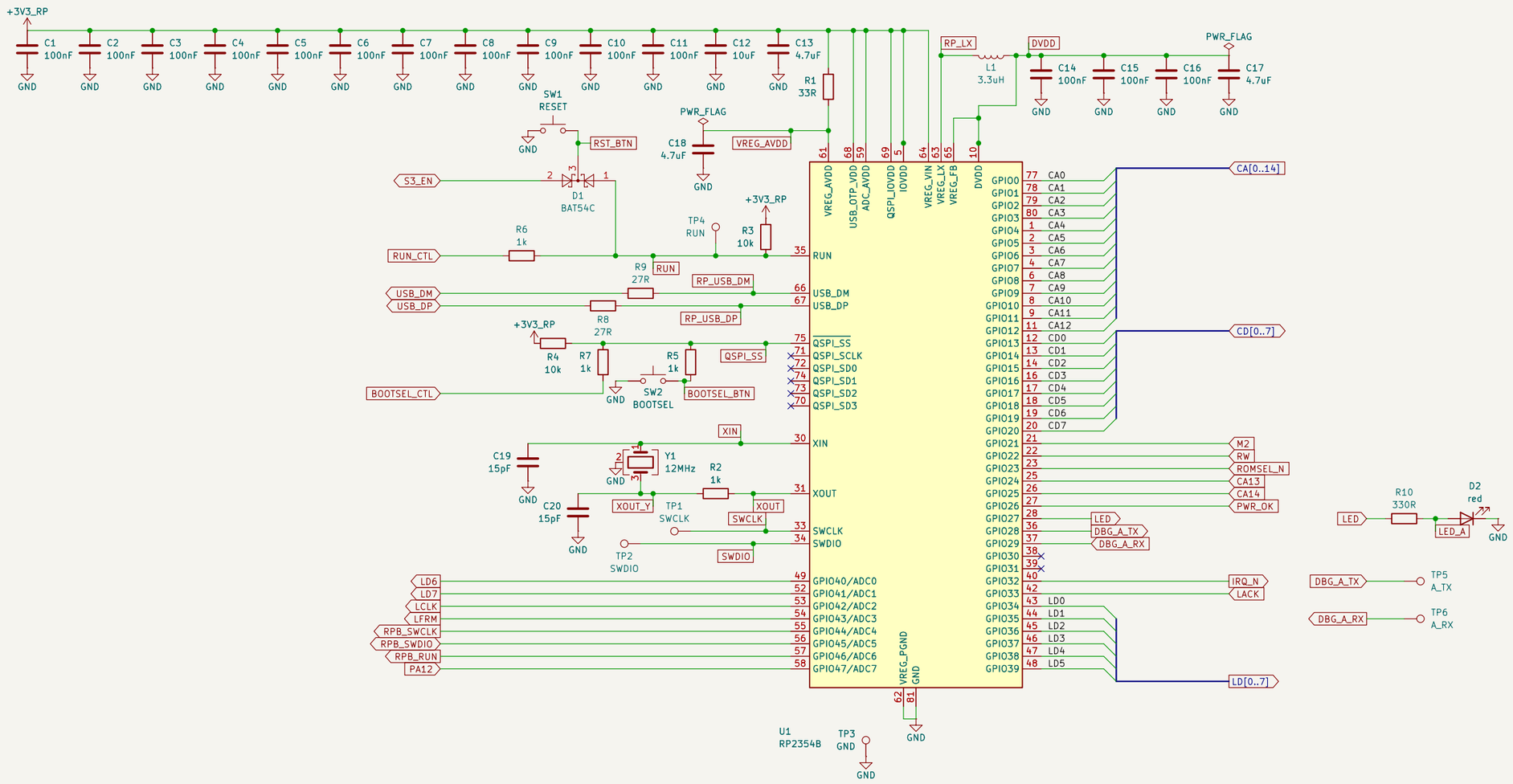
Size: A3	Date: 2026–10–18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/7

NES 72-pin edge drawn by function (finger numbers on the pins). Pin 1 is at the RIGHT seen
from the label side with the fingers down (nesdev: NES-EWROM-01 geometry).
EXP0-9, SYSTEM CLK and PPU /A13 are not used. The CIClone (ATtiny13A) is DNP until its firmware exists.



Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CART)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file
FujiNet

Sheet: /edge-cic/		
File: edge-cic.kicad_sch		
Title: FujiNet NES Rev1 — 72-pin cart edge, CIClone		
Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 2/7



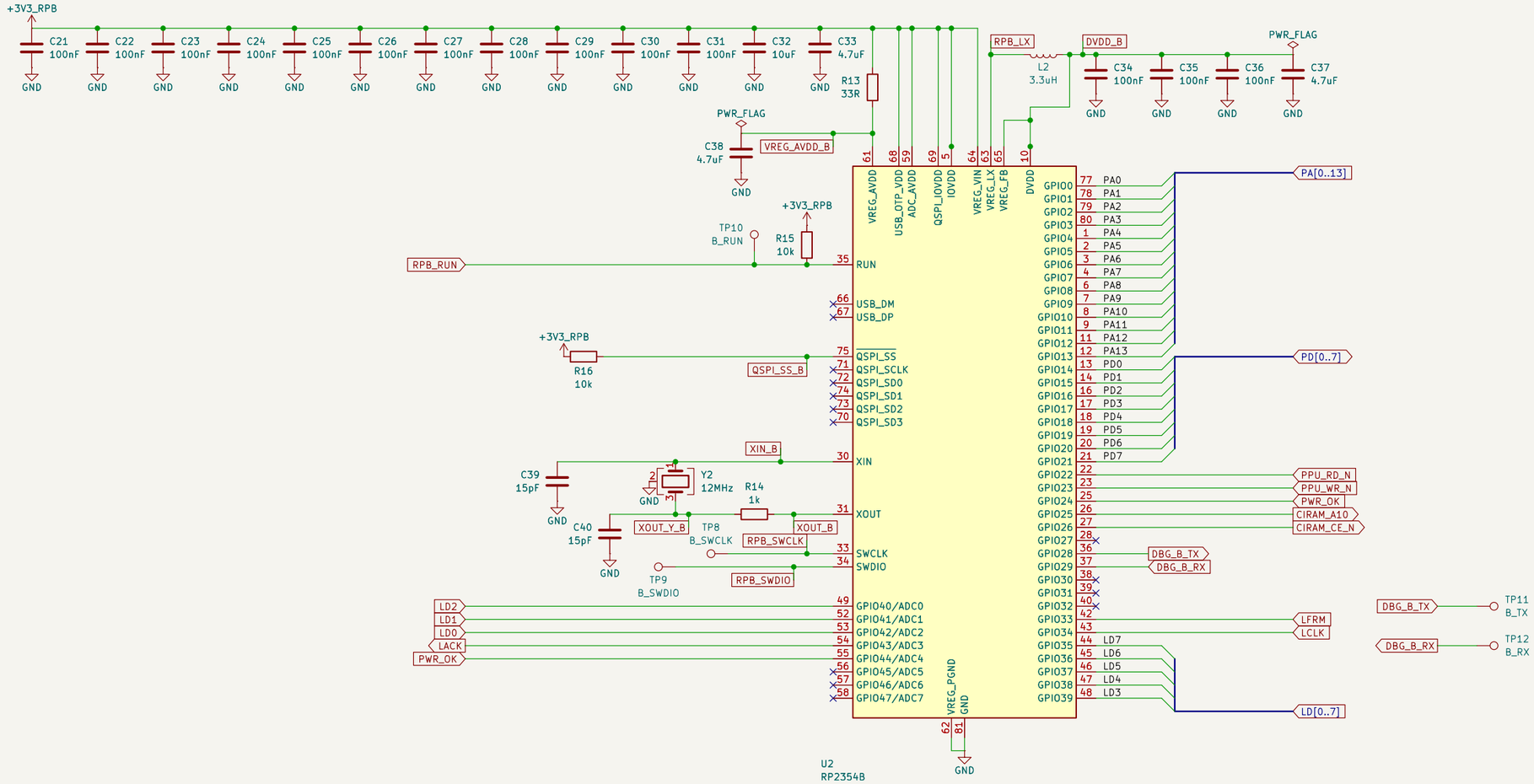
Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CART)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /rp-cpu/
File: rp-cpu.kicad_sch

Title: FujiNet NES Rev1 — RP-A: CPU bus, PRG, link, RP-B loader

Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/7



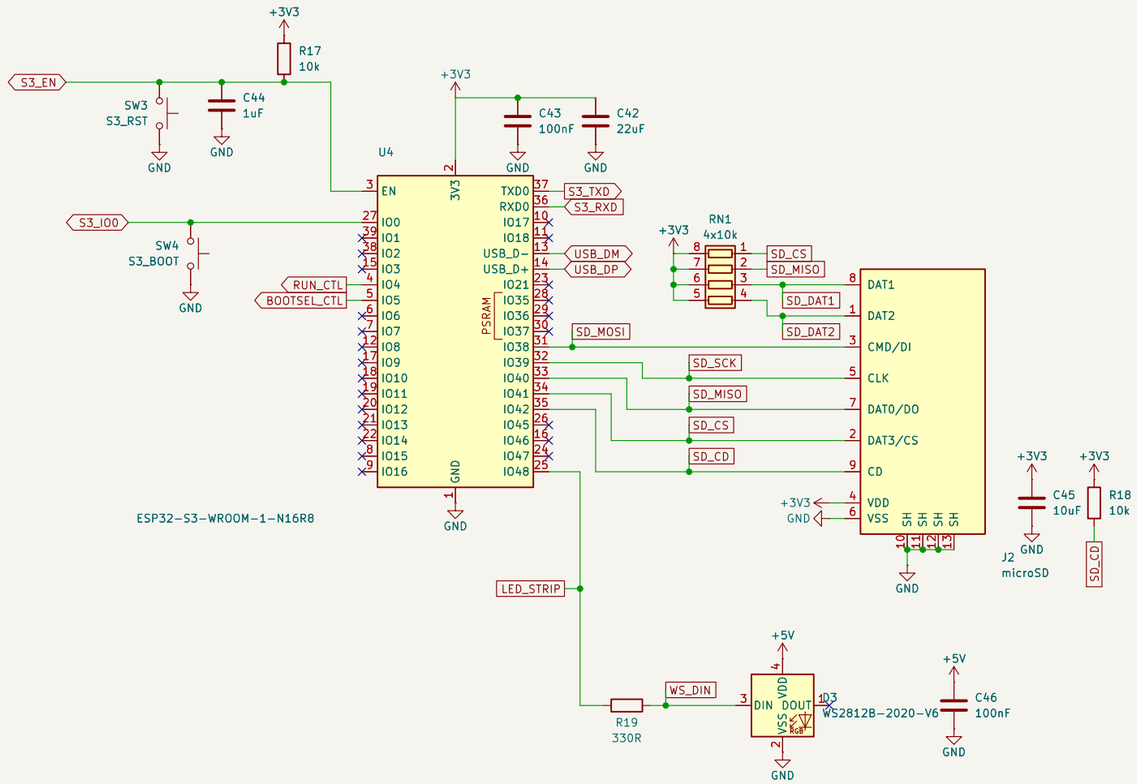
Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /rp-ppu/
File: rp-ppu.kicad_sch

Title: FujiNet NES Rev1 — RP-B: PPU bus, CHR and CIRAM

Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/7



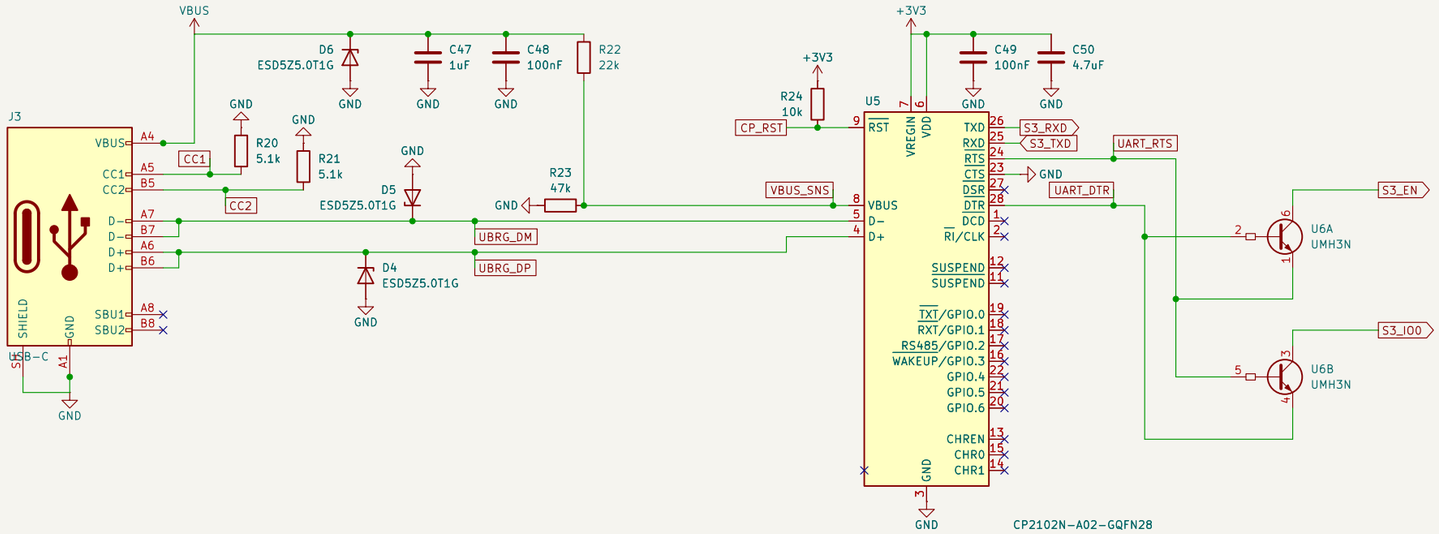
Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /esp32s3-sd/
File: esp32s3-sd.kicad_sch

Title: FujiNet NES Rev1 — ESP32-S3, microSD, status LED

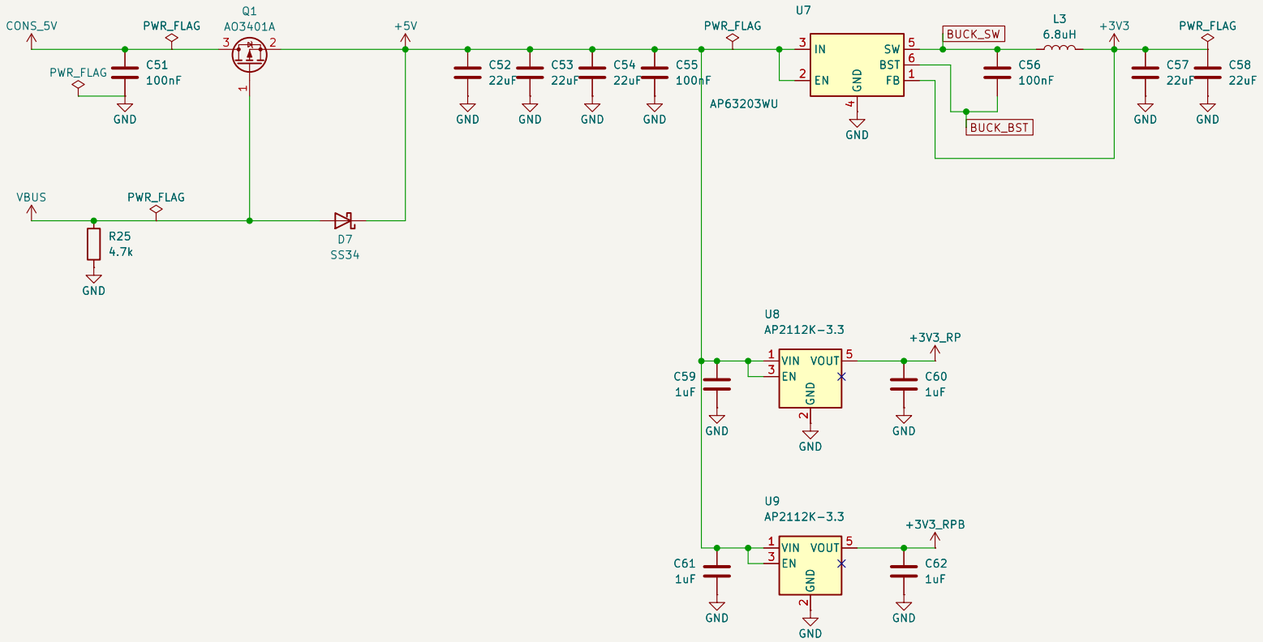
Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/7



Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /usb-uart/		
File: usb-uart.kicad_sch		
Title: FujiNet NES Rev1 — USB-C, CP2102N bridge		
Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 6/7



Rev1: two RP2354Bs, no SRAM; see docs/design-review-rev1.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet NES Rev1 — Power: 5V OR, 3.3V buck, two RP LDOs

Size: A3	Date: 2026-10-18	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 7/7